

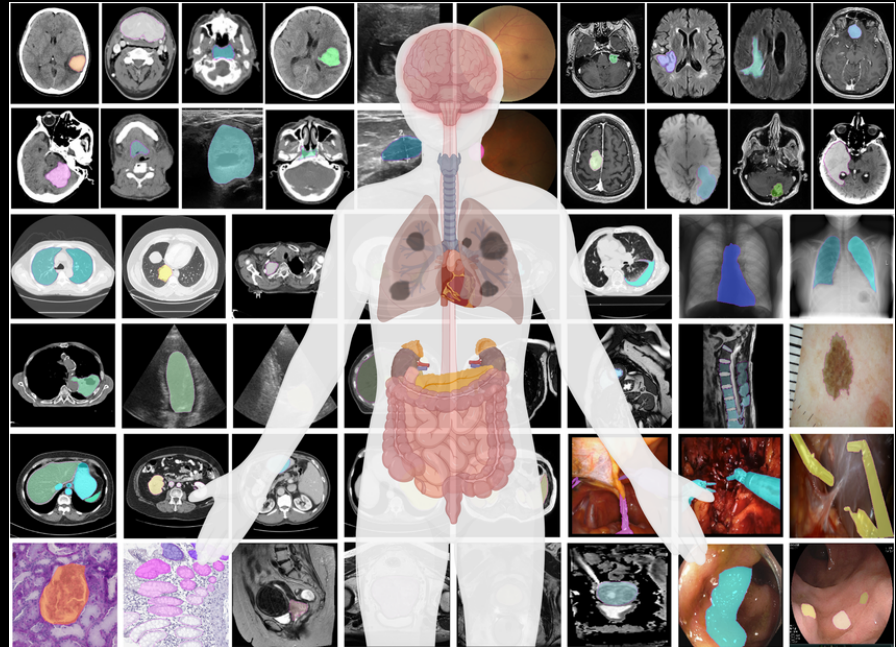
LECTURE 2: COMMON TASKS

WHAT WE ASK OF A MODEL, AND WHY THE DATA FIGHTS BACK

Mehmet Kurt, Tianyi Ren
University of Washington

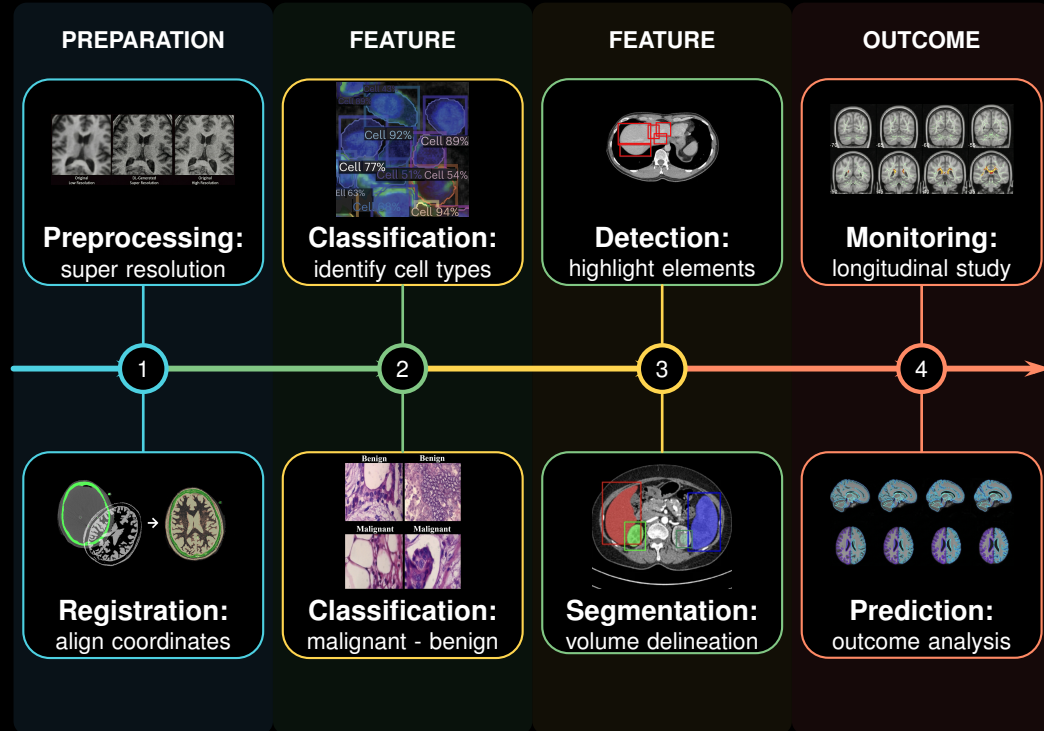
August 20, 2026

COMMON TASKS IN MEDICAL IMAGE ANALYSIS

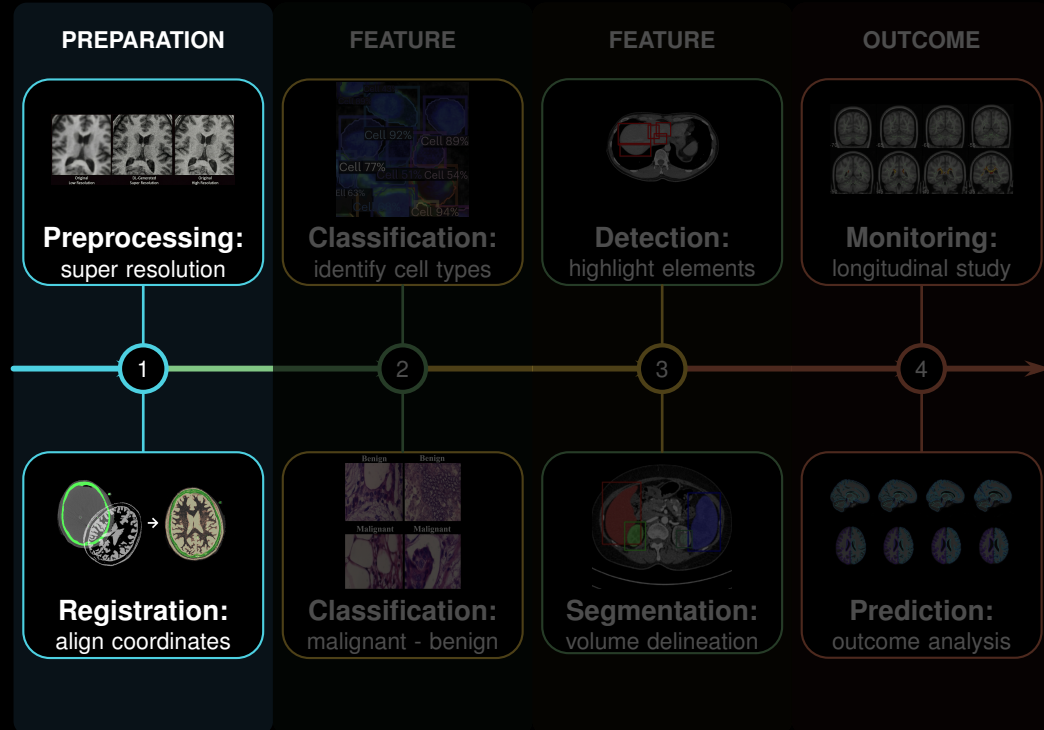


Source: Wang et al., Nature Communications 2024

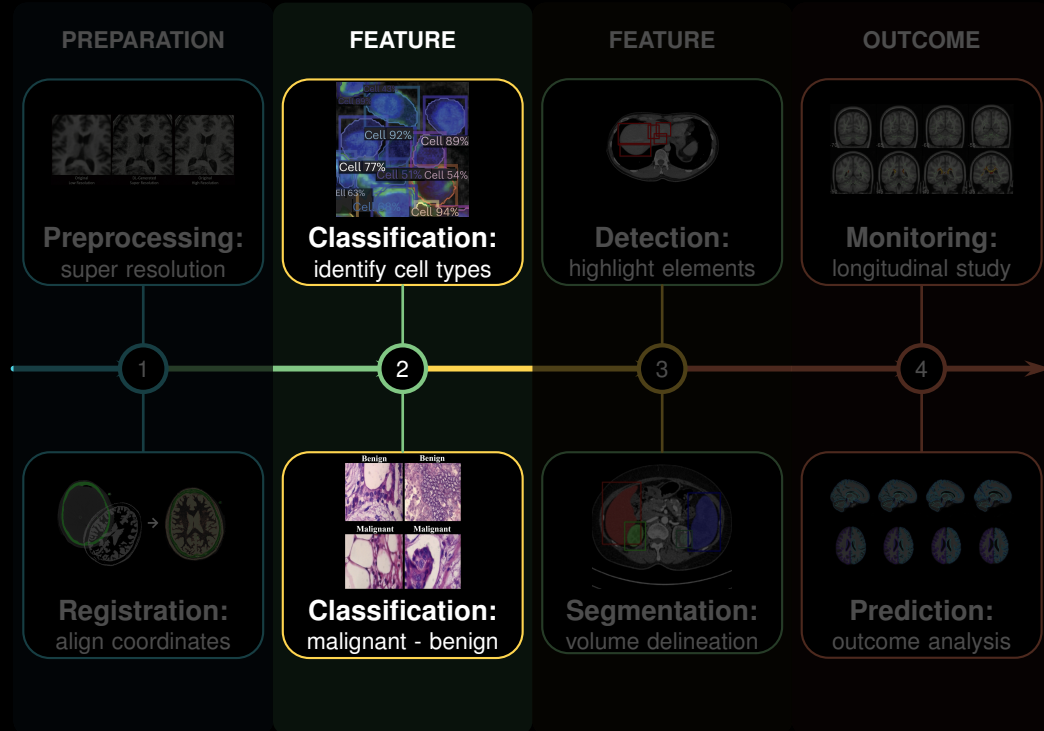
COMMON TASKS IN MEDICAL IMAGE ANALYSIS



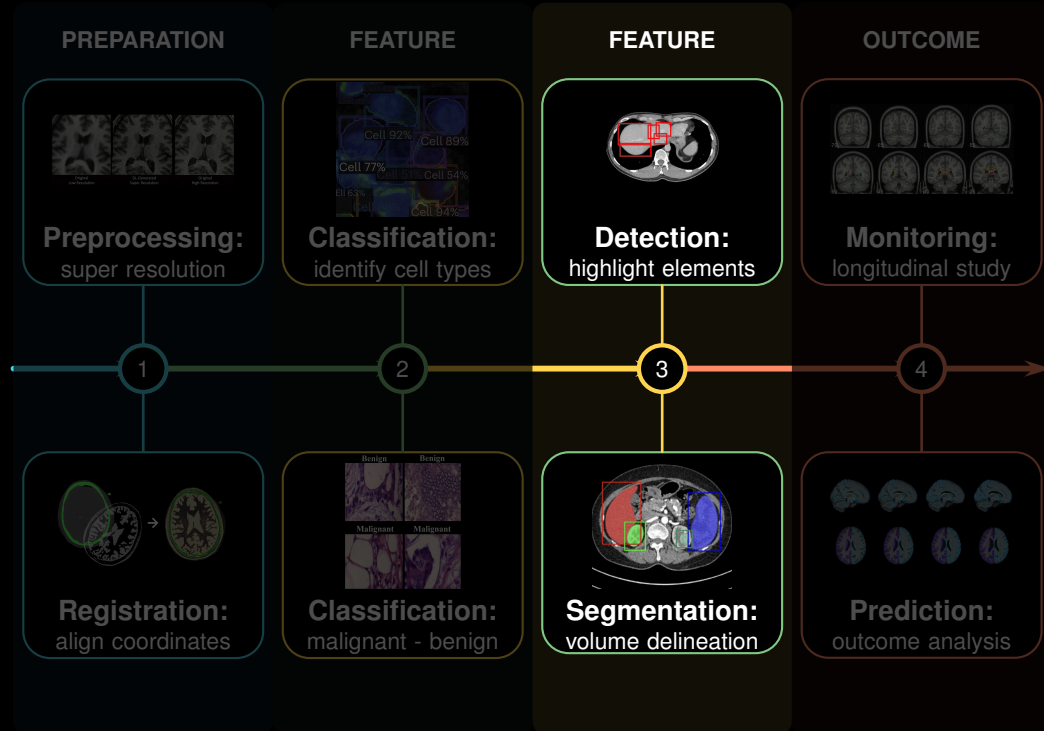
COMMON TASKS IN MEDICAL IMAGE ANALYSIS



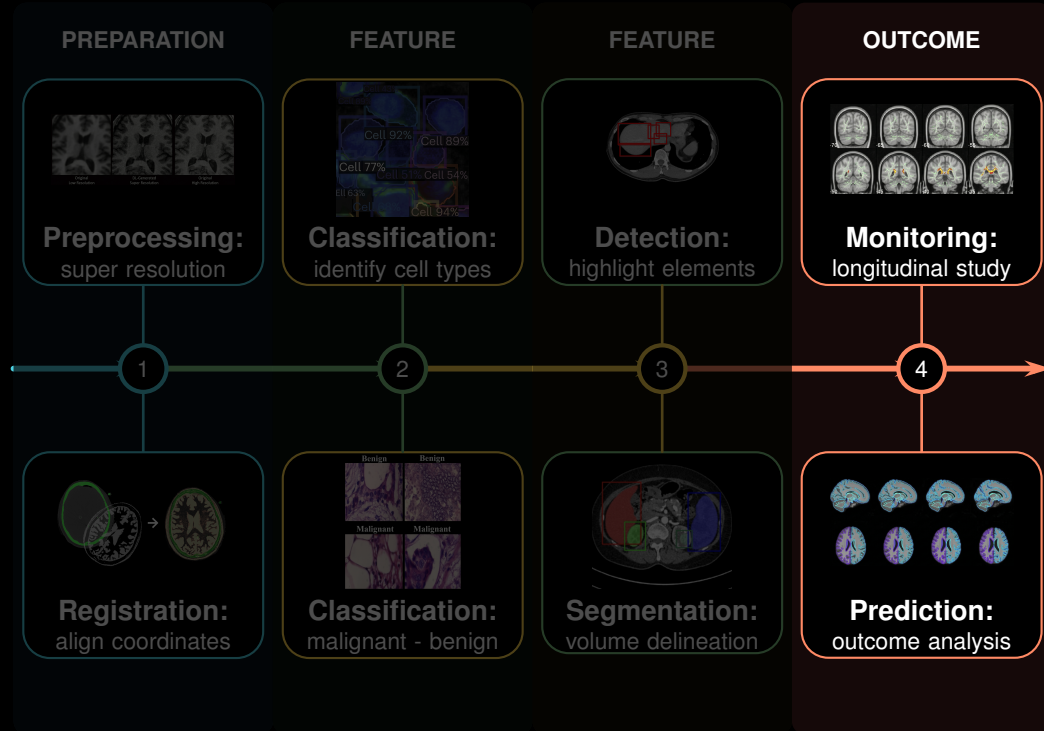
COMMON TASKS IN MEDICAL IMAGE ANALYSIS



COMMON TASKS IN MEDICAL IMAGE ANALYSIS



COMMON TASKS IN MEDICAL IMAGE ANALYSIS

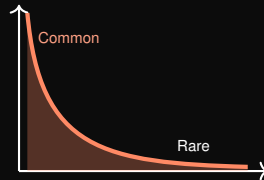


MEDICAL IMAGE DATA IS INTRINSICALLY CHALLENGING

Data Scarcity

Modalities: high-res 3D volumes but few patient samples.

Long-tail: common diseases dominate; rare cases are hard to obtain.



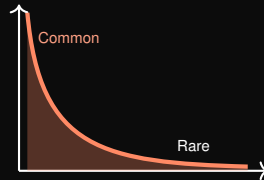
Imbalance: common vs. rare diseases

MEDICAL IMAGE DATA IS INTRINSICALLY CHALLENGING

Data Scarcity

Modalities: high-res 3D volumes but few patient samples.

Long-tail: common diseases dominate; rare cases are hard to obtain.



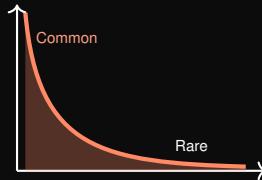
Imbalance: common vs. rare diseases

MEDICAL IMAGE DATA IS INTRINSICALLY CHALLENGING

Data Scarcity

Modalities: high-res 3D volumes but few patient samples.

Long-tail: common diseases dominate; rare cases are hard to obtain.



Imbalance: common vs. rare diseases

Distribution Shift

Non-standard: vendors (Siemens vs. GE), protocols differ.

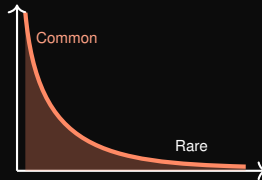
Subjective labels: inter-observer variability.

MEDICAL IMAGE DATA IS INTRINSICALLY CHALLENGING

Data Scarcity

Modalities: high-res 3D volumes but few patient samples.

Long-tail: common diseases dominate; rare cases are hard to obtain.



Imbalance: common vs. rare diseases

Distribution Shift

Non-standard: vendors (Siemens vs. GE), protocols differ.

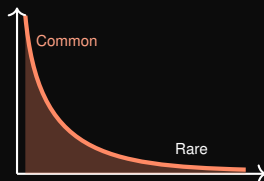
Subjective labels: inter-observer variability.

MEDICAL IMAGE DATA IS INTRINSICALLY CHALLENGING

Data Scarcity

Modalities: high-res 3D volumes but few patient samples.

Long-tail: common diseases dominate; rare cases are hard to obtain.



Imbalance: common vs. rare diseases

Distribution Shift

Non-standard: vendors (Siemens vs. GE), protocols differ.

Subjective labels: inter-observer variability.

Privacy Constraints

De-identification risk (e.g., facial reconstruction).

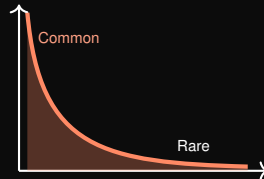
Siloed data due to HIPAA / GDPR constraints.

MEDICAL IMAGE DATA IS INTRINSICALLY CHALLENGING

Data Scarcity

Modalities: high-res 3D volumes but few patient samples.

Long-tail: common diseases dominate; rare cases are hard to obtain.



Imbalance: common vs. rare diseases

Distribution Shift

Non-standard: vendors (Siemens vs. GE), protocols differ.

Subjective labels: inter-observer variability.

Privacy Constraints

De-identification risk (e.g., facial reconstruction).

Siloed data due to HIPAA / GDPR constraints.